

Learning the metric induced by a G_2 -structure

Generalisation and equivariance through rotation augmentation

Eduardo Toffolo

University of Campinas

MM845 — AI for Geometry

Code and reproduction instructions: <https://github.com/EToffolo/AI-Project>

Abstract. A positive 3-form in seven dimensions determines a unique metric through a known algebraic formula. This project tests how accurately a generic neural regressor learns that map and whether rotating both inputs and targets improves its geometric behaviour. Exact synthetic labels allow separate tests of prediction, equivariance and homogeneity. With 21,000 training examples, rotation augmentation reduces the mean seed-wise median relative error from 25.81% to 24.99% in distribution and from 43.57% to 42.79% under extrapolation. Rotation defects also decrease, but remain substantial. The results support a modest benefit from augmentation while showing the limits of learning mathematical structure without imposing it.

1 Introduction and related work

A G_2 -structure encodes a metric using a differential 3-form. Its pointwise linear algebra is explained by Karigiannis [1]; Grigorian [2] gives explicit metric formulas in the study of deformations. Here the problem is restricted to one oriented vector space. There are no differential equations or torsion-free conditions. The known formula makes this a controlled benchmark for learning a nonlinear tensor-valued map.

Symmetry provides a natural inductive bias. Cohen and Welling [3] construct layers that enforce group equivariance, whereas Chen, Dobriban and Lee [4] interpret augmentation as averaging over group orbits and analyse variance reduction. Their results motivate the experiment; they do not guarantee equivariance of the generic regressor used here. The question is whether $SO(7)$ augmentation reduces rotation defects without materially degrading accuracy, under a fixed training protocol.

2 Mathematical background

Let $V = \mathbb{R}^7$ with dual basis $e^1, \dots, e^7$, orientation $\text{vol}_0 = e^1 \wedge \dots \wedge e^7$, and $e^{ijk} = e^i \wedge e^j \wedge e^k$. Use the reference form

$$\varphi_0 = e^{123} + e^{145} + e^{167} + e^{246} - e^{257} - e^{347} - e^{356}. \quad (1)$$

The positive orbit is $\mathcal{P} = \{A^*\varphi_0 : A \in GL^+(7)\}$, where $(A^*\varphi)(u, v, w) = \varphi(Au, Av, Aw)$ and $GL^+(7)$ denotes positive-determinant invertible matrices. Its stabiliser is $G_2 \subset SO(7)$, so $F : \mathcal{P} \rightarrow \text{Sym}^+(7)$ is well defined by $F(A^*\varphi_0) = A^\top A$. $\text{Sym}^+(7)$ is the set of symmetric positive-definite 7×7 matrices.

In this orientation convention [2], the exact formula is

$$B_{ij}(\varphi) \text{vol}_0 = \frac{1}{6} (\iota_{e_i} \varphi) \wedge (\iota_{e_j} \varphi) \wedge \varphi, \quad F(\varphi) = (\det B)^{-1/9} B, \quad (2)$$

where ι_{e_i} contracts a form with e_i . Indeed, $B = g_\varphi \sqrt{\det g_\varphi}$ implies $\det B = (\det g_\varphi)^{9/2}$. Naturality and cubic dependence of B give the identities

$$F(C^*\varphi) = C^\top F(\varphi) C \quad (C \in GL^+(7)), \quad F(t\varphi) = t^{2/3} F(\varphi) \quad (t > 0). \quad (3)$$

These identities are diagnostics for the learned map, rather than constraints built into its architecture. In particular, the target transforms under rotations; it is not an invariant scalar label.

3 Data generation and experimental design

Draw independent Haar rotations $Q_1, Q_2 \in \text{SO}(7)$ using Gaussian QR with sign and determinant corrections, and independent log singular values s_i . Generate

$$A = Q_1 \text{diag}(e^{s_1}, \dots, e^{s_7}) Q_2, \quad (\varphi, g) = (A^* \varphi_0, A^\top A). \quad (4)$$

The learner receives only the 35 coefficients φ_{ijk} , $i < j < k$, in lexicographic order; no $3!$ rescaling is used. It never receives A or s . The eigenvalues of g are e^{2s_i} . Thus generation guarantees positivity and provides exact labels independently of the exterior-algebra implementation of Eq. (2).

For the in-distribution (ID) population, $s_i \sim U[-0.35, 0.35]$. A single dataset of 30,000 pairs, generated with seed 2026, is randomly split into 21,000 training, 4,500 validation and 4,500 test sources. A separate out-of-distribution (OOD) test contains 4,500 pairs with $s_i \sim U[-0.7, 0.7]$, conditioned on $\max_i |s_i| > 0.35$. This shifts singular values beyond the training range, rather than merely choosing new rotations. Metric eigenvalues lie in approximately $[0.497, 2.014]$ for ID and $[0.247, 4.055]$ for OOD; the corresponding upper bounds on the spectral condition number are $e^{1.4} \approx 4.06$ and $e^{2.8} \approx 16.44$.

Splitting precedes augmentation. Transformed copies inherit their source split, preventing leakage between training and held-out data. Inputs are standardised coefficient by coefficient using training-only means and standard deviations; per-sample norms are retained. Nested training subsets of 2,100, 7,000 and 21,000 examples are shared by all methods within each seed (11, 22 and 33). At the largest size, the three seeds use the same full training set. Validation and test sets are fixed, so seed variation measures training variation, not uncertainty over new datasets.

The source population is already rotation invariant: replacing A by AQ leaves its distribution unchanged because Q_2Q remains Haar. Augmentation therefore provides new orientations of a finite sample, without expanding population support. Float64 label checks compare Eq. (2), using only φ , with $A^\top A$ on 64 randomly selected ID and 64 OOD pairs. The maximum relative discrepancy is 5.61×10^{-16} ; $F(\varphi_0) = I$ is also verified.

4 Learning methods

Ridge baseline. An affine map predicts the 28 independent entries of a symmetric matrix, without enforcing positivity. It minimises squared errors of those entries plus an ℓ_2 penalty on weights. An unpenalised intercept is included. The penalty is chosen from $\{10^{-4}, 10^{-2}, 1, 10^2, 10^4\}$ using validation relative loss.

Neural regressors. The architecture is a 35–128–128–28 multilayer perceptron (MLP) with two ReLU hidden layers. Its outputs form a lower triangular matrix L : off-diagonal entries are unrestricted, and diagonal entries are $\text{softplus}(z) + 10^{-5}$. The prediction is $\hat{F}(\varphi) = LL^\top$, ensuring symmetry and positivity in exact arithmetic. Both MLPs minimise

$$\mathcal{L} = \frac{1}{m} \sum_{a=1}^m \frac{\|\hat{F}(\varphi_a) - g_a\|_F^2}{\|g_a\|_F^2}, \quad (5)$$

with Adam [5], learning rate 10^{-3} and batch size 256. The maximum is 100 epochs; early stopping uses patience 15 and improvement threshold 10^{-5} . The checkpoint with the smallest validation loss is restored.

The augmented MLP draws a fresh independent Haar rotation per example per minibatch and trains on $(Q^* \varphi, Q^\top g Q)$, before input standardisation. Initial weights and minibatch orders match the ordinary MLP within each seed; rotation randomness uses a separate stream. Maximum epochs and optimiser settings match, although actual epochs and computation may differ. All 27 fitted models (9 ridge, 18 MLP) run on CPU. Test data do not select hyperparameters or checkpoints.

5 Results and analysis

5.1 Predictive accuracy and sample size

For each held-out pair define the relative Frobenius error

$$E(\varphi, g) = \frac{\|\hat{F}(\varphi) - g\|_F}{\|g\|_F}, \quad \|M\|_F^2 = \sum_{i,j} M_{ij}^2. \quad (6)$$

This measures error of the entire matrix, counting off-diagonal entries twice. A 25% error is not a classification failure rate. Unlike the training loss in Eq. (5), the statistics below are not squared.

Table 1: Predictive errors (%) at $n = 21,000$. Medians: mean $\pm$ sample standard deviation over three seeds. p95: mean of seed-wise 95th percentiles.

Model	ID: 4,500 examples		OOD: 4,500 examples	
	Median	p95	Median	p95
Ridge	36.58 ± 0.00	48.29	62.29 ± 0.00	72.86
MLP	25.81 ± 0.09	33.92	43.57 ± 0.08	64.60
MLP + rotations	24.99 ± 0.07	33.25	42.79 ± 0.01	62.87

The MLP’s median error is about 29% lower than ridge on ID and 30% lower on OOD. Augmentation adds a smaller gain: 3.18% relative to the plain MLP on ID and 1.78% on OOD, or about 0.82 and 0.78 percentage points. Each seed improves, but errors remain high.

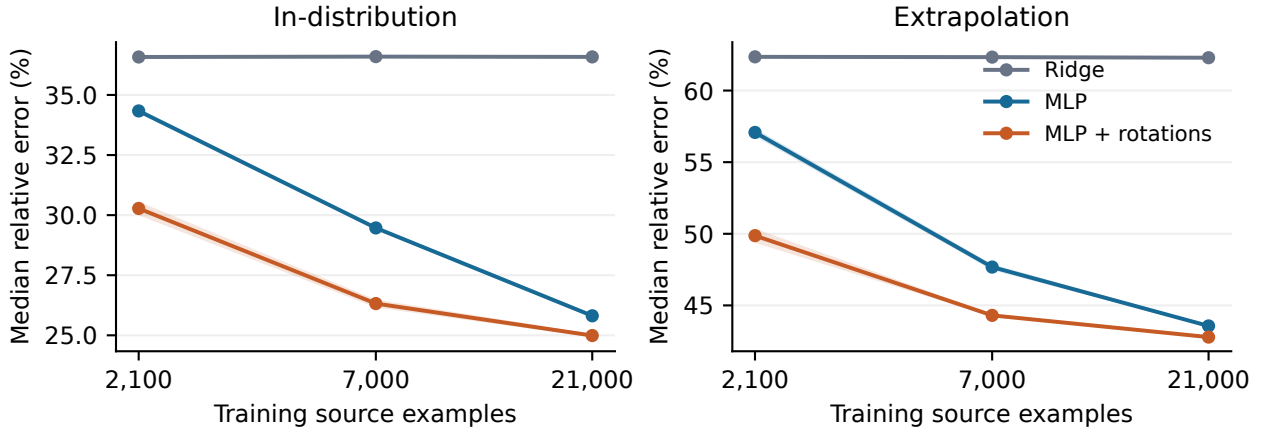


Figure 1: Median predictive error versus source sample size. Lines average seed-wise medians; shading is $\pm$ one sample standard deviation, not a confidence interval.

Larger datasets improve the neural regressors (Figure 1), while ridge stays close to 36.6% ID and 62.3% OOD. At 2,100 sources, augmentation changes the mean median from 34.34% to 30.28% ID and from 57.08% to 49.87% OOD: relative gains of approximately 11.8% and 12.6%. The declining gain suggests augmentation is most useful when finite samples cover orientations sparsely; the curves do not isolate that explanation causally.

Extrapolation remains much harder. For the augmented MLP, the OOD median is 42.79%, versus 24.99% ID, and the OOD p95 reaches 62.87%. Rotations preserve the singular values of A and cannot directly expose the learner to the broader OOD singular-value range. The gain therefore does not establish reliable extrapolation to arbitrary scales or conditioning.

All methods have zero non-positive-definite predictions on the base ID and OOD tests: at the largest size, zero failures in 13,500 predictions per method and domain across seeds. Ridge has no positivity guarantee; the MLP guarantee holds in exact arithmetic, so numerical failures are explicitly counted.

5.2 Equivariance and homogeneity

On 512 fixed randomly selected sources per domain, evaluate the covariance defect for fresh transformations C :

$$D_C(\varphi, g) = \frac{\|\hat{F}(C^*\varphi) - C^\top \hat{F}(\varphi) C\|_F}{\|C^\top g C\|_F}. \quad (7)$$

The denominator uses the exact transformed metric. Rotations are Haar draws; $GL^+(7)$ transformations use Eq. (4) with log bound 0.2. Indices and transformations are shared across methods and seeds. Transformed inputs may leave their source domain’s singular-value range.

Table 2: Geometric defects at $n = 21,000$: mean $\pm$ sample standard deviation of seed-wise medians. All values are dimensionless.

Model	ID sources		OOD sources	
	SO(7)	$GL^+(7)$	SO(7)	$GL^+(7)$
Ridge	0.0152 $\pm$ 0.0000	0.2123 $\pm$ 0.0000	0.0132 $\pm$ 0.0000	0.1459 $\pm$ 0.0000
MLP	0.2355 $\pm$ 0.0016	0.2658 $\pm$ 0.0029	0.2985 $\pm$ 0.0057	0.3244 $\pm$ 0.0027
MLP + rotations	0.2223 $\pm$ 0.0021	0.2519 $\pm$ 0.0028	0.2817 $\pm$ 0.0078	0.3118 $\pm$ 0.0044

Augmentation reduces the mean rotation median by 5.61% ID and 5.61% OOD. The declared criterion requires a smaller rotation median in each seed and domain, allowing at most a 5% relative increase in predictive median at the largest size. All six comparisons satisfy it, with improved accuracy. $GL^+(7)$ defects also decline. Three seeds and shared tests do not establish statistical significance or exact equivariance.

Ridge illustrates a pitfall: for seed 11, its mean ID prediction is close to $1.08288I$, with only about 1.12% mean relative variation. A constant map $\hat{F}(\varphi) = cI$ has zero rotation defect, since $Q^\top(cI)Q = cI$, while predicting the wrong metric. Accuracy and covariance must be assessed together.

For $t \in \{0.25, 0.5, 2, 4\}$, define a homogeneity defect and a scaled prediction error:

$$H_t = \frac{\|\hat{F}(t\varphi) - t^{2/3}\hat{F}(\varphi)\|_F}{\|t^{2/3}g\|_F}, \quad E_t = \frac{\|\hat{F}(t\varphi) - t^{2/3}g\|_F}{\|t^{2/3}g\|_F}. \quad (8)$$

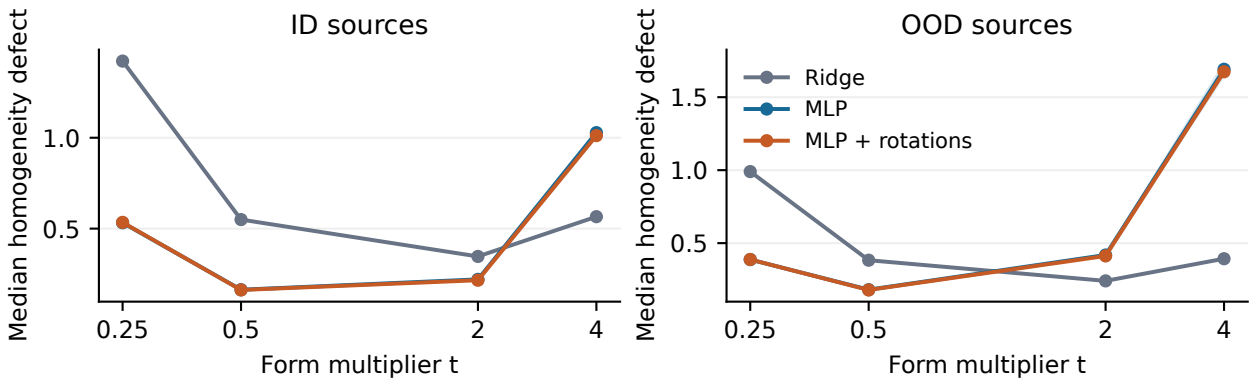


Figure 2: Homogeneity defects on 512 sources per domain, aggregated as in Figure 1. Nearly overlapping MLP curves show little effect of rotations on scaling.

At $t = 4$, augmentation changes mean median E_t from 104.9% to 103.0% ID and from 172.3% to 168.2% OOD. Both MLPs fail severely; rotations do not enforce the degree-2/3 law.

6 Discussion, conclusion and extensions

Why are the errors still large? Labels are exact, so the discrepancy comes from approximation, optimisation and generalisation. The map involves cubic contractions followed by determinant normalisation, beyond the expressive scope of an affine regressor. The MLP is a generic approximator with no built-in homogeneity or equivariance. More data help, but the experiment does not isolate whether capacity, optimisation budget or coverage of the 35-dimensional input domain is the main bottleneck.

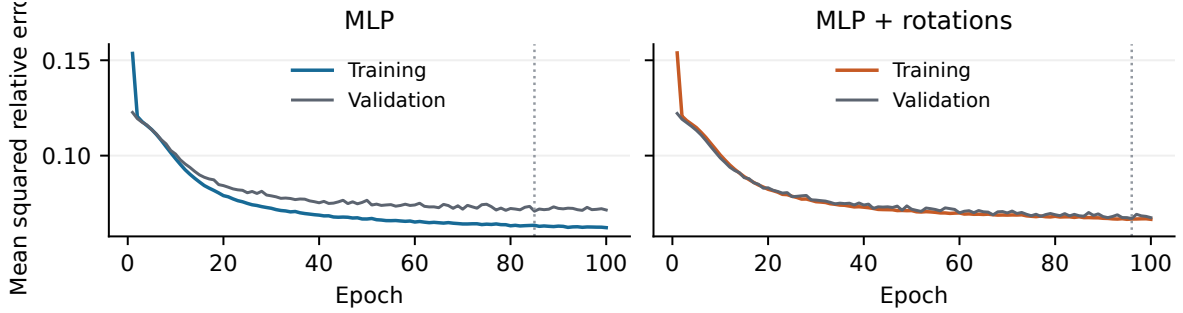


Figure 3: Training and validation loss for 21,000 sources, seed 11. Dotted lines mark selected checkpoints. Augmented training uses changing rotated samples; validation always uses the original fixed sources.

The nearby training and validation curves (Figure 3) suggest no large ID generalisation gap. Augmented models select epochs 96, 98 and 97, near the maximum of 100: compatible with limited capacity or insufficient optimisation, without proving either. Mean measured training time is 18.9 seconds for the plain MLP and 39.3 seconds with augmentation on the recorded CPU environment. Equal maximum epochs do not mean equal time; early stopping may give different update budgets. No speed comparison with the exact formula was performed.

Conclusions. Neural models are more accurate than ridge, and rotation augmentation consistently improves prediction and rotation defects under this protocol. The benefit is strongest at smaller sample sizes and modest at the largest size. Median errors near 25% ID and 43% OOD and substantial geometric defects prevent treating the learned map as a precise replacement for the formula. Positivity, accuracy, equivariance and homogeneity are separate properties: observing one does not establish the others. The scope is a bounded pointwise benchmark; global manifolds, torsion-free structures and near-singular limits remain unexplored.

Extensions. First test whether the network can deliberately overfit 128–256 sources; then compare larger architectures and longer training with validation-only selection. An exact homogeneity constraint is possible: for $r = \|\varphi\|_2 > 0$, use $\hat{F}(\varphi) = r^{2/3}h(\varphi/r)$. This preserves scale while making the identity automatic. A further comparison should use an $SO(7)$ -equivariant tensor architecture or features based on the contractions defining B . To study a broader range of singular values, expand training support and reserve a new external range as an independent OOD test. These are proposed experiments, not results obtained here; observed test scores should not guide retrospective hyperparameter selection.

Reproducibility and use of AI. The repository linked on page 1 includes environment instructions and `configs/full.json`. From its root, after activating the documented environment:

```
python -m g2metric run --config configs/full.json --output outputs/repro
```

A new output directory regenerates data, all 27 fits, metrics, figures and checkpoints. Raw run outputs are excluded from Git; this report includes the central results. Saved manifests record seeds, versions and hashes. All 24 automated tests and the smoke pipeline pass. Codex assisted implementation, experiments, analysis and drafting. Checks include independent exterior-algebra labels, pullback identities, source separation and checkpoint persistence. The author remains responsible for reviewing the submission.

References

- [1] S. Karigiannis. *Introduction to G_2 geometry*. Lecture notes, 2019; revised 2020. [arXiv:1909.09717](#).
- [2] S. Grigorian. *Deformations of G_2 -structures with torsion*. *Asian Journal of Mathematics*, 20(1):123–155, 2016.
[arXiv:1108.2465](#); [doi:10.4310/AJM.2016.v20.n1.a6](#).
- [3] T. S. Cohen and M. Welling. *Group equivariant convolutional networks*. Proceedings of the 33rd International Conference on Machine Learning, PMLR 48:2990–2999, 2016. <https://proceedings.mlr.press/v48/cohenc16.html>.
- [4] S. Chen, E. Dobriban and J. H. Lee. *A group-theoretic framework for data augmentation*. Journal of Machine Learning Research, 21(245):1–71, 2020. <https://jmlr.org/papers/v21/20-163.html>.
- [5] D. P. Kingma and J. Ba. *Adam: A method for stochastic optimization*. International Conference on Learning Representations, 2015. [arXiv:1412.6980](#).